

NAME: GALLA MOHAN SAI

MODULE NO : 3

Exercise 1: Control Structures:

Sample Example 1:

Scenario 1: Apply 1% Discount on Loan Interest Rate for Customers Above 60
DECLARE

```
CURSOR c_customers IS  
  SELECT CustomerID  
  FROM Customers  
  WHERE Age > 60;
```

BEGIN

```
FOR cust IN c_customers LOOP
```

```
  UPDATE Loans  
  SET InterestRate = InterestRate - 1  
  WHERE CustomerID = cust.CustomerID;
```

```
  DBMS_OUTPUT.PUT_LINE(  
    '1% Interest Rate Discount Applied to Customer ID: '  
    || cust.CustomerID  
  );
```

```
END LOOP;
```

```
COMMIT;
```

```
END;
```

```
/
```

EXAMPLE OUTPUT :

```
1% Interest Rate Discount Applied to Customer ID: 101  
1% Interest Rate Discount Applied to Customer ID: 104  
1% Interest Rate Discount Applied to Customer ID: 109
```

Scenario Example 2 : Set VIP Status for Customers with Balance Over \$10,000

DECLARE

```
CURSOR c_customers IS
  SELECT CustomerID
  FROM Customers
  WHERE Balance > 10000;
```

BEGIN

```
FOR cust IN c_customers LOOP
```

```
  UPDATE Customers
  SET IsVIP = 'TRUE'
  WHERE CustomerID = cust.CustomerID;
```

```
  DBMS_OUTPUT.PUT_LINE(
    'VIP Status Assigned to Customer ID: '
    || cust.CustomerID
  );
```

```
END LOOP;
```

```
COMMIT;
```

```
END;
```

```
/
```

OUTPUT

VIP Status Assigned to Customer ID: 102

VIP Status Assigned to Customer ID: 105

VIP Status Assigned to Customer ID: 108

Scenario Example 3:

Send Reminders for Loans Due Within Next 30 Days DECLARE

DECLARE

```

CURSOR c_loans IS
    SELECT c.CustomerID,
           c.Name,
           l.LoanID,
           l.DueDate
    FROM Customers c
    JOIN Loans l
    ON c.CustomerID = l.CustomerID
    WHERE l.DueDate BETWEEN SYSDATE AND SYSDATE + 30;

```

```

BEGIN

```

```

    FOR loan_rec IN c_loans LOOP

```

```

        DBMS_OUTPUT.PUT_LINE(
            'Reminder: Dear ' || loan_rec.Name ||
            ', your Loan ID ' || loan_rec.LoanID ||
            ' is due on ' ||
            TO_CHAR(loan_rec.DueDate, 'DD-MON-YYYY')
        );

```

```

    END LOOP;

```

```

END;

```

```

/

```

OUTPUT:

Reminder: Dear Mohan, your Loan ID 1001 is due on 15-JUL-2026

Reminder: Dear Ravi, your Loan ID 1003 is due on 20-JUL-2026

Exercise 3: Stored Procedures

Scenario Example 1 : Process Monthly Interest for Savings Accounts

Stored Procedure: ProcessMonthlyInterest:

```

CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest

```

```

IS

```

```

BEGIN

```

```

    UPDATE Accounts
    SET Balance = Balance + (Balance * 0.01)
    WHERE AccountType = 'Savings';

```

```
COMMIT;
```

```
DBMS_OUTPUT.PUT_LINE(  
    'Monthly Interest Processed Successfully.'  
);
```

```
END;
```

```
/
```

Execute Procedure:

```
BEGIN
```

```
    ProcessMonthlyInterest;
```

```
END;
```

```
/
```

Output:

Monthly Interest Processed Successfully.

Scenario Example 2 : Update Employee Bonus

Stored Procedure: UpdateEmployeeBonus

```
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus(  
  
    p_DepartmentID IN NUMBER,  
    p_BonusPercent IN NUMBER  
  
)  
IS  
BEGIN
```

```
    UPDATE Employees
```

```
    SET Salary = Salary + (Salary * p_BonusPercent / 100)
```

```
    WHERE DepartmentID = p_DepartmentID;
```

```
COMMIT;
```

```
DBMS_OUTPUT.PUT_LINE(  
    'Employee Bonus Updated Successfully.'  
);
```

```
END;
```

```
/
```

Execute Procedure:

```
BEGIN
    UpdateEmployeeBonus(101, 10);
END;
/
```

Output

Employee Bonus Updated Successfully.

Sample Example 3 :**Transfer Books Between Libraries**

```
CREATE OR REPLACE PROCEDURE TransferFunds(
```

```
    p_FromAccount IN NUMBER,
    p_ToAccount   IN NUMBER,
    p_Amount      IN NUMBER
```

```
)
IS
```

```
    v_Balance NUMBER;
```

```
BEGIN
```

```
-- Get the balance of the sender's account
```

```
SELECT Balance
```

```
INTO v_Balance
```

```
FROM Accounts
```

```
WHERE AccountID = p_FromAccount;
```

```
-- Check whether sufficient balance is available
```

```
IF v_Balance >= p_Amount THEN
```

```
    -- Deduct amount from sender's account
```

```
    UPDATE Accounts
```

```
    SET Balance = Balance - p_Amount
```

```
    WHERE AccountID = p_FromAccount;
```

```
-- Add amount to receiver's account
```

```
UPDATE Accounts
```

```

SET Balance = Balance + p_Amount
WHERE AccountID = p_ToAccount;

COMMIT;

DBMS_OUTPUT.PUT_LINE('Transfer Successful.');
```

ELSE

```

DBMS_OUTPUT.PUT_LINE('Insufficient Balance.');
```

END IF;

EXCEPTION

```

WHEN NO_DATA_FOUND THEN
    DBMS_OUTPUT.PUT_LINE('Account Not Found.');
```

```

WHEN OTHERS THEN
    DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);
```

END;

/ Execute :

```

BEGIN
    TransferFunds(1004, 1008, 3000);
END;
```

/

Sample Output (Successful Transfer)

Transfer Successful.

Sample Output (Insufficient Balance)

Insufficient Balance.

Sample Output (Invalid Account)

Account Not Found